

process and use the Micro USB male to iPhone male OTG connector.

Do check all connections to avoid damaging your expensive phone. After connecting the OTG

and sensor with the Raspberry Pi, fix the sensor, Raspberry Pi, and OTG adaptor inside the designed and 3D-printed enclosure (see Fig. 5, Fig. 6, and Fig. 7).

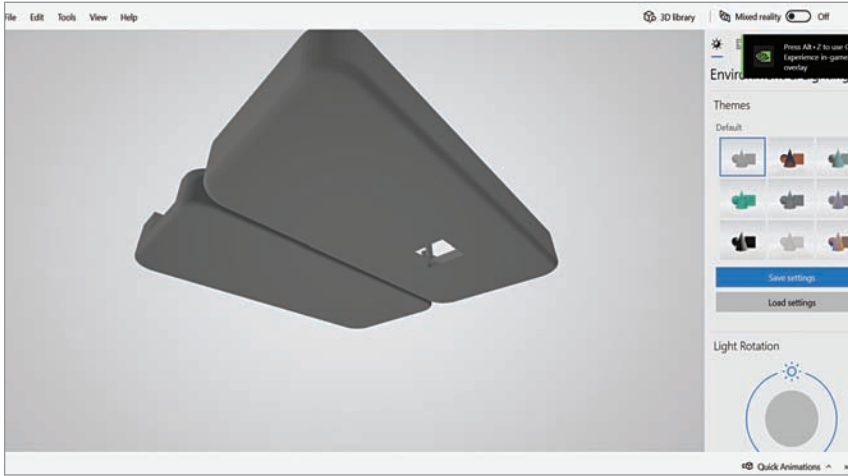


Fig. 3: The device case design

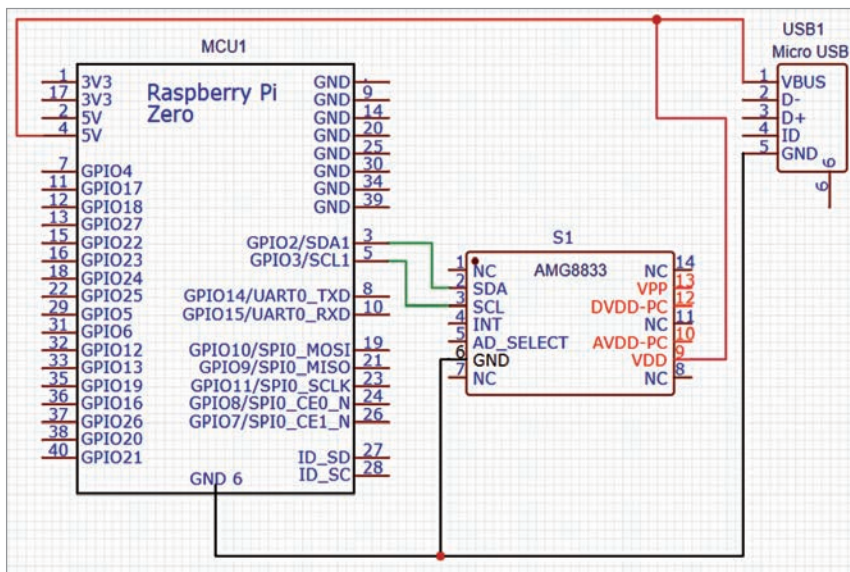


Fig. 4: Circuit diagram

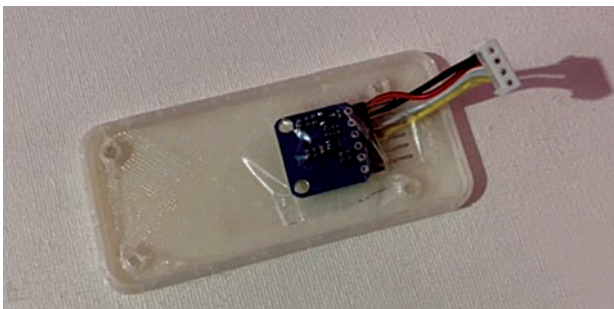


Fig. 5: Thermal sensor attached to the designed case



Fig. 6: Raspberry Pi fixed on the 3D-printed case

Coding

Install the Raspberry Pi with the latest Raspbian OS. Then enable I2C peripheral in Raspberry Pi, as the thermal sensor we are using is interfaced on I2C communication. Hence, to enable the I2C on Raspberry Pi, open terminal and run `sudo raspi-config` and then go to advance and interfacing option and enable the I2C and VNC.

Now, connect the Raspberry Pi to a phone's hotspot or Wi-Fi network to which the phone is connected. Next, install a few libraries and modules that help us with interfacing the sensor and getting data from it, as well as viewing that data in video form. So open the Linux terminal and run the following command:

```
sudo pip3 install pygame
sudo pip3 install scipy
sudo pip3 install numpy
sudo pip3 install Adafruit_AMG88xx
```

After installing the libraries, import all installed modules in the code. Then set the maximum and minimum temperature range to get the AMG sensor readings. Set the range between 22 and 45. Even a higher range can be set, depending on the requirement and the purpose for which you are using it. If it is to be used for thermal analysis and the temperature range is greater than 100, you can set the range accordingly.

Next, set the pygame window size, the size at which the video output for your thermal camera is obtained. Then map the sensor pixel data to red, green, blue, and yellow.

Let us understand how the thermal camera works. The thermal